Journal of Communications Research
Volume 6, Number 2
ISSN: 1935-3537
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GAMING FOR GOOD: VIDEO GAMES AND ENHANCING PROSOCIAL BEHAVIOUR

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ABSTRACT

The number of publications pertaining to video gaming and its effects on subsequent behavior has more than tripled from the past to the current decade. This surge of research parallels the ubiquitousness of video game play in everyday life, and the increasing concern of parents, educators, and the public regarding possible deleterious effects of gaming. Numerous studies have now investigated this concern. Recently, research has also begun to explore the possible benefits of gaming, in particular, increasing prosocial behaviour. This chapter presents a comprehensive review of the research literature examining the effects of video game playing on prosocial behaviour.

Within this literature, a variety of theoretical perspectives and research methodologies have been adopted. For example, many researchers invoke the General Learning Model to explain the mechanisms by which video games may influence behaviour. Other researchers refer to frameworks involving moral education, character education, and care-ethics in their examination of the relationship between gaming and prosocial development. Diverse parameters have been explored in these studies. For example, different studies have assessed both the immediate and delayed impacts of gaming, and investigated the effects of different durations of video game playing. Additionally, based on each study's operational definitions of "aggressive behaviour" and "prosocial behaviour", a variety of behaviours have been assessed and different measures have been employed. For instance, studies have used self-report measures of empathy, the character strengths of generosity and kindness, and the level of civic engagement, as well as used word-completion and story completion tasks and tit-for-tat social situation games such as "Prisoners' Dilemma". These studies have examined changes in both aggressive and prosocial behaviours.

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"Video games" collectively span a wide spectrum of content, contexts, and player engagement modalities. Therefore, researchers have focused on different aspects of the video games and on different aspects of the context or structure within which the games are played. Competitive versus cooperative games have been studied, as have antisocial versus prosocial games; additionally, combinations of these facets (e.g., games that use violent or antisocial methods to achieve prosocial goals) have been studied. The effects of solo game playing versus co-playing with another research participant have also been studied.

A cross-section of the variety of perspectives, methodologies, findings and foci of research within the study of video game playing and prosocial behaviour is included in this literature review. Despite the wide array of research approaches and questions, gaps exist within the video game—prosocial behaviour literature. This is not surprising, given that this is a relatively new domain of research. We conclude the chapter, therefore, by proposing future research questions and directions with which to address this gap in our knowledge.

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Typical impressions regarding video games, the effects of playing video games, and of gamers themselves, are largely negative. These impressions are oversimplified at best, and simply wrong, at worst. One reason for this negative impression, is the portrayal of gamers by the mainstream media as lonely individuals who play video games secluded in their basements (Gamespot, 2013). This is a gross misportrayal. In reality, most video game playing occurs within a social context—either with friends or family in the same room, or interacting with other players via a range of massively multiplayer on-line (MMO) games (Entertainment Software Association [ESA], 2013).

Also contributing to the negative impressions surrounding gaming and gamers is that the media is quick to create headlines out of tenuous links between tragic events and video game playing. For example, both print and televised media reports of the horrific shootings on the Virginia Tech campus in 2013 blamed video games as the impetus for the shootings. However, a police search of the perpetrator's dorm room found no evidence that he played video games, and his university roommate stated that he had never seen the perpetrator play video games (Benedetti, 2013). Conversely, the media is largely silent on the many prosocial, cooperative acts of generosity that large groups of gamers perform. For example, in 2013, approximately 500 gamers from all over the country brought their laptops to Portland, Oregon to participate in a gaming party weekend. These gamers also brought 37,500 pounds of food which they donated to the Union Gospel Mission, a local charity that helps people in need. Only one, small, local news station reported on the event (Steelman, 2013). This was not a rare occurrence of gamer generosity; there are many regular "gaming for good" charity events (Manuel, 2012), but positive news stories about playing video games and gamers receive little, if any, media attention.

Contributing to the negative impression of gaming and gamers, is that the vast majority of research examining the effects of playing video games has focused on the negative effects of playing violent video games (e.g., Anderson et al., 2010; Barlett, Anderson, & Swing, 2009; Griffiths, 1999; Sherry, 2001). This is unrepresentative on two levels: firstly, by its sole

focus on negative effects (analogous to traditional psychology's focus on ill-being), and secondly, by its focus mainly on *violent* video games.

However, the perspective of positive psychology, which examines well-being, is beginning to be seen in video game research. Researchers have begun to study the positive effects, and the wide range of possible benefits, of playing video games (e.g., Allaire et al., 2013; Granic, Lobel, & Engels, 2014; Johnson, Jones, Scholes, & Carras, 2013; Khoo, 2012). Additionally, nonviolent, and even explicitly prosocial, video games are now the focus of studies (e.g.,Greitemeyer & Osswald, 2010; Greitemeyer & Osswald, 2011; Rosenberg, Baughman, & Bailenson, 2013; Saleem, Anderson, & Gentile, 2012). Moreover, as researchers develop a more sophisticated understanding of the negative *and* positive effects of video games, it is becoming evident that video games cannot be neatly assigned to mutually exclusive categories of "violent" and "nonviolent" or "prosocial" and "antisocial" (Ferguson & Garza, 2011; Gentile et al., 2009; Tear & Nielsen, 2013). Many games combine elements of each of these categories, such as games in which players engage in violent acts for prosocial reasons or to achieve prosocial "good for all" goals. Likewise, video games are not always easily dichotomized into "competitive" or "cooperative" games (Eastin, 2007; Wadley, Gibbs, Hew, & Graham, 2003). Many competitive games contain cooperative elements within them, such as when players form "guilds" that work as cohesive units in competing against evil forces.

Our intent in this chapter is not to negate the decades of high-quality research which has demonstrated negative effects of playing violent video games. We acknowledge the importance of these findings, particularly in light of the ubiquitousness of video game play in every day life and the increasing amount of violent content found in different forms of entertainment media today (e.g., television, movies, internet sites), and in many popular video games.

Rather, our purpose here is to highlight the positive effects of playing video games by presenting a selection of the extant research in this area. We seek to broaden readers' perspectives of what constitutes a "video game" and to deepen readers' understanding of the variety of content found in video games (i.e., nonviolent, prosocial, neutral, violent, or mixed), the context within which these games are played (e.g., cooperative, competitive, first-person, third-person, solo, in a group), and how content and context can, singly and via an interaction, moderate or differentially effect prosocial behaviour subsequent to playing the video game. We also seek to explore the various methodologies employed in researching the beneficial effects of playing video games, as well as the theories of mechanisms by which these effects may occur. As with any new research field, gaps exist within the prosocial gaming literature. Therefore, we will conclude the chapter by outlining future research questions and directions.

WHAT CONSTITUTES A "VIDEO GAME"?

Although first-person action games, electronic role-playing games, participation in a virtual reality world, electronic puzzle-based or arcade "pinball" type games, and interactive novels each have their own defining features as games, they share two properties: all are interactive activities wherein the players' actions influence the outcome or scenarios of the

game; and, all are, of course, mediated via a computer interface. Like Koo & Seider (2010), "we are less interested in drawing lines around what constitutes a 'video game' than we are in considering the possibilities that video games and video game-like experiences offer to prosocial learning" (p. 17). Therefore, for the purposes of this chapter, we consider *all* of the above examples to fall within the domain of video games. As David Cage, creator of *Beyond* (an interactive drama action-adventure video game), said, "A video game can be so many different things. … Let's open this medium to whoever has different ideas" (quoted in Williams, 2013).

PROSOCIAL GAMES RESULT IN PROSOCIAL BEHAVIOUR: CONVERGENT EVIDENCE

Akin to the broad spectrum of video games that exists, a broad spectrum of ideas exists regarding how to define and measure prosocial, beneficial effects of playing video games. The use of diverse methodologies (e.g., correlational, experimental, longitudinal), operational definitions (ranging from a reduction in aggressive behaviours and thoughts to an increase in prosocial behaviours and thoughts), and measures (e.g., self-report measures of mood and empathy, behavioural measures of generosity and cooperation, and word-completion and story-completion measures to assess access to cognitive schemas), have led to a sizable body of literature illuminating the many prosocial effects that can result from playing video games. Moreover, these effects have been studied using numerous video games (see Appendix for a list and brief description of all video games used in studies highlighted in this chapter).

Indeed, many studies, using different age groups from various countries, have identified the beneficial effects of playing prosocial video games. In this section, we highlight a representative sampling of these studies. Participants in these studies range in age from 9 to 61 years, and come from countries with markedly different cultures—the United States, Germany, Singapore, and Japan.

For example, Gentile et al. (2009) surveyed 727 Singaporean 7th and 8th grade children. Reflective of the population of Singapore, a mix of ethnicities were represented, including Chinese, Malay, and Indian. Participants were asked to list "their three favourite video games, estimate the number the number of hours per week spent playing each game, rate how often players help others in the game, and how often players hurt or kill others in the game" (p. 755). Participants also completed several self-report measures assessing different aspects of prosocial behaviour such as helping behaviour, cooperation and sharing, empathic attitudes, and emotional awareness. Self-report measures of aggressive tendencies were also administered, examining levels of aggressive cognitions and approval of aggression. Hostile attribution bias was measured via six story-completion tasks in which the child chose an explanation for a character's actions described in various ambiguous provocative situations. Even after controlling for total amount of time gaming, violent game exposure, sex, and age, prosocial game exposure was positively related to all of the prosocial behaviours and traits measured. Furthermore, exposure to prosocial video games was negatively related to all measures of aggressive behaviour, thus suggesting a dual effect of enhancing prosocial behaviour while reducing aggressive or antisocial behaviour.

To help identify the causal relation, Gentile et al. (2009) conducted an experiment. Undergraduates from a college in the United States (N = 161) were randomly assigned to play either a neutral (*Pure Pinball* or *Super Monkey Ball Deluxe)*, prosocial (*Super Mario Sunshine* or *Chibi Robo*), or violent (*Ty2* or *Crash Twinsanity*) video game for 20 minutes. Participants then assigned to their partner 11 puzzles, which, if completed within 10 minutes, would win their partner a $10 gift certificate. Participants had a choice of 30 puzzles to choose from, which ranged in difficulty from easy to hard. The number of easy and the number of difficult puzzles that participants chose for their partners to complete were used as an assessment of helping (prosocial) or hurting (aggressive) behaviour. Participants who had played a prosocial video game were significantly more helpful than were participants who had played either a neutral or violent video game. Moreover, choosing easy puzzles to help one's partner was positively related to the amount of prosocial content in the assigned game.

In a related study (Greitemeyer, Agthe, Turner, and Gschwendtner, 2012), before playing either a neutral (*Pinball*), prosocial (*Firefighters: Saving Lives*), or violent (*Mortal Kombat*) video game, participants (66 undergraduates from a university in Germany) wrote an essay that another participant would evaluate and comment on. In turn, participants reviewed and commented on their (nonexistent) partner's essay. All of the "feedback" that the actual participants received was negative, and concluded with a written comment on the bottom that the essay was boring. After participants played the video game they had been randomly assigned for 15 minutes, they were told that the study was over. However, participants were then asked if they would take part in an unrelated study involving the effects of aversive learning on memory in which the person who had evaluated their essay was participating. Participants were told that this individual, who had given the participant such a bad evaluation on their essay, would be paid according to how well he or she did in the memory tasks. It was explained that the participant's task was to choose the level of aversive noise that this (in reality, non-existent) individual would hear when they made an error. Higher volumes of aversive noise chosen by participants indicated higher levels of retaliation and aggression. Consistent with results reported by Gentile et al. (2009), compared to participants who had played either the neutral or the violent video game, participants who had played the prosocial video game chose significantly lower levels of aversive noise, thereby indicating lower levels of aggressive behaviour. Moreover, these results remained significant even after controlling for levels of trait altruism, aggression, arousal, and mood.

The effects of playing a prosocial video game on positive mood have also been studied. For example, Whitaker and Bushman (2012) examined not only the effects of playing video games on prosocial behaviour, but also the effects on positive mood. In this experiment, 116 undergraduates were randomly assigned to play either a neutral (*Super Mario Galaxy*, *Wii Sports Resort*), relaxing (*Endless Ocean*, the fishing mini-game in *The Legend of Zelda*, *Twilight Princess*), or violent (*Resident Evil 4*, *No More Heroes*) video game. After playing, participants completed a self-report measure of positive affect. Participants were then told that the study was over, but were asked if they could stay to help the researcher prepare for another study by sharpening some pencils. Compared to participants who had played either the neutral or the violent video game, participants who had played the relaxing video game reported greater positive affect and displayed greater helping behaviour, as measured by the number of pencils they sharpened to help the researcher. Additionally, there was a mediation effect of positive mood, whereby the resulting positive mood produced from playing a

relaxing video game significantly increased the amount of help that the participants exhibited after they thought the study was over.

To test whether playing prosocial video games would yield similar beneficial effects in a younger sample, Saleem, Anderson, and Gentile (2012) examined the differential effects of playing either a neutral (*Pure Pinball*, *Super Monkey Ball Deluxe*), prosocial (*Chibi Robo*), or violent (*Ty2*, *Crash Twinsanity*) video game in 191 children ranging in age from 9 to 14 years. Each child played a randomly assigned video game for 30 minutes, after which each child chose 11 puzzles for an ostensible partner to complete. Participants were informed that if their partner completed 10 of the 11 puzzles, within 10 minutes, then their partner would receive a $10 gift certificate. Hurting (aggressive) or helping (prosocial) behaviour was thus assessed by how difficult or easy participants made it for their "partners" to complete the puzzles and win the gift certificate—that is, by how many difficult puzzles and how many easy puzzles the participant chose for their partner to complete. As in Whitaker & Bushman's (2012) study, those who had played the prosocial video game chose significantly less difficult puzzles for their partner to complete than did those who had played either the neutral or the violent video game.

This pattern of increased helping behaviour after playing a prosocial video game has been consistently demonstrated. For example, Greitemeyer and Osswald (2010) conducted a series of four experiments. In Experiment 1, after playing 8 minutes of either a neutral (*Tetris*), prosocial (*Lemmings*), or violent video game (*Lamers*), participants (N = 54) who had played the prosocial video game were more helpful in assisting the researcher pick up a collection of dropped pencils than were participants who had played either the neutral or violent video game. A different measure of helping behaviour was used in Experiment 2 (N = 40) (i.e., participant agreement to assist in further studies and the amount of time the participant agreed to devote to this). All of the participants who had played the prosocial video game agreed to help in further studies, whereas only 68% of those who had played the neutral game agreed to help. Moreover, of the participants who were willing to help, participants who had played the prosocial game agreed to devote significantly more time than did participants who had played the neutral game.

In Experiment 3 (N = 36), Greitemeyer and Osswald (2010) added a second prosocial game (*City Crisis*) to determine the generalizability of their findings to other prosocial games. They also utilized a complex prosocial behaviour with associated high costs as an indicator of prosocial behaviour (i.e., intervening on behalf of a passive individual who was being harassed by an aggressive abuser). This abusive interaction was, of course, staged, but participants believed it to be authentic. Again, playing a prosocial video game facilitated helping behaviour; significantly more participants intervened who had played either of the two prosocial games than did those who had played the neutral video game.

In addition to the effects on mood and behaviour, the effects that playing prosocial video games have on thoughts and accessibility to either prosocial or aggressive cognitions have also been studied. The results of several studies support a relationship between playing prosocial video games and an increase in subsequent prosocial thoughts. When behaviour was also studied as a dependent variable in these studies, an increase in prosocial thoughts led to an increase in prosocial behaviour.

The fourth experiment (N = 37) in Greitemeyer and Osswald's (2010) series, replicated the results from Experiment 1 by again utilizing the helping behaviour measure of picking up dropped pencils, in addition to examining the extent to which playing a prosocial video game

would increase prosocial thoughts. After playing either the neutral or prosocial video game, participants wrote down what they had been thinking about while playing the game. These lists of thoughts were rated and coded as either prosocial or neutral (no aggressive thoughts had been listed). Participants who had played the prosocial video game reported significantly more prosocial thoughts than did participants who had played the neutral video game. Furthermore, there was a significant mediation effect of the accessibility of prosocial thoughts (measured by the number of prosocial thoughts listed) on the positive relationship between playing a prosocial (compared to neutral) video game and subsequent helping behaviour (measured by the number of dropped pencils that participants picked up). These results extended previous findings concerning affect as a mediator of the relationship between playing a prosocial video game and subsequent prosocial behaviour (Whitaker & Bushman, 2012).

Greitemeyer and Osswald (2011) have also used a lexical decision task as a measure of accessibility to prosocial social thoughts. After playing a neutral (*Tetris*) or prosocial (*Lemmings*) video game for 10 minutes, participants (N = 47 adults ranging in age from 17 to 61 years) engaged in a lexical decision task which involved deciding if a string of letters displayed on the screen in front of them was a word or not. In cases in which the letters spelled a word, half of the words were prosocial (e.g., help) and half were neutral (e.g., run). Reaction times to specific categories of words are one measure of the accessibility of thoughts pertaining to that category—the faster the reaction time, the more accessible that category of thoughts is to the participant engaged in the lexical decision task. Reaction times to neutral target words and non-words did not significantly differ between the two groups. However, compared to participants who had played the neutral video game, participants who had played the prosocial video game responded significantly faster to prosocial target words.

Evidence for a relationship between playing prosocial video games and a subsequent decrease in accessibility to aggressive cognitions has also been demonstrated. In one study (Greitemeyer & Osswald, 2009), 48 undergraduates completed three ambiguous story stems after playing either a neutral (*Tetris*) or prosocial (*Lemmings*) video game for 10 minutes. Paralleling the results of studies measuring access to prosocial cognitions after playing a prosocial video game, participants who had played the prosocial video game expected fewer aggressive responses from the characters in the stories, and were less likely to expect the characters to say, do, think, or feel something aggressive than were participants who had played the neutral video game. Similar results were found for undergraduates who engaged in a word-completion task after playing either a neutral or prosocial video game. Compared to participants who had played a neutral video game, participants who had played a prosocial game had lower antisocial word completion scores.

Parallel to studies which demonstrated that an increase in prosocial thoughts leads to an increase in prosocial behaviour, a decrease in accessibility to aggressive thoughts has been demonstrated to lead to a decrease in aggressive behaviour. For example, in a follow-up study to their aversive noise experiment described previously, Greitemeyer et al. (2012) measured not only the level of aggressive behaviour subsequent to playing either a neutral (*Tetris*) or prosocial (*Lemmings*) video game, but also access to aggressive cognitions. Prior to playing the video game, participants answered four questions from a visual analogy task. Participants were given only 90 seconds to answer these questions in order to induce self-doubt about their performance. Participants then played their randomly assigned video game for 12 minutes, after which they received (false) negative feedback pertaining to their "abysmal"

performance on the prior visual analogy task. State hostility was then measured using a validated self-report instrument, and a word-completion task was administered to assess accessibility of aggressive thoughts. Items (e.g., HA_E) could be completed to spell either an aggressive word (e.g., HATE) or a neutral word (e.g., HAVE). Greitemeyer et al. then used the aversive noise measure of aggression; participants were given the opportunity to choose the level of aversive noise that the individual who had given them the negative feedback would receive in a subsequent (purportedly, unrelated) experiment.

Results were consistent with Greitemeyer et al.'s (2012) previous findings and with Whitaker and Bushman's (2012) findings: playing a prosocial video game (compared to playing a neutral video game) resulted in participants exhibiting lower levels of aggression. Compared to participants who had played the neutral video game, not only did participants who had played the prosocial video game choose lower levels of aversive noise to be delivered, these participants also reported lower levels of state hostility, and demonstrated lower accessibility to aggressive cognitions as measured by their word-completion responses. Furthermore, a mediation effect of both affect and cognition on the relationship between playing prosocial video games and lowered aggressive behaviour was found, thus replicating previous findings (Greitemeyer & Osswald, 2010; Whitaker & Bushman, 2012).

Studies we have highlighted have examined prosocial effects exhibited shortly after playing prosocial video games. Results of two studies, both conducted in Japan, suggest that these prosocial effects are enduring. Ihori, Sakamoto, Shibuya, and Yukawa (2007) conducted two surveys of 5th grade elementary schoolchildren (N = 780 across the two surveys) from eight public schools (four urban and four rural). The follow-up survey was administered three months after the initial survey was completed. Children were asked to report on their video game playing habits (e.g., how many hours in a day did they play video games), and to rate the frequency with which they saw video game scenes of violence and prosociality. The children also ranked their preference of video games from lists which included games from genres such as violent, realistic sports, puzzle-based, and music-based, and completed a self-report measure assessing frequency of their aggressive and prosocial behaviour for the last month. Sample items included “I punched or kicked others” and “I was kind to other people” (p. 171).

Ihori et al. (2007) used a cross-lagged effect structural equation model analysis to examine causal relationships between type of video game play and prosocial effects over the three-month period. Results suggested that, for both boys and girls, increased exposure to prosocial video game scenes resulted in a subsequent increase of prosocial behaviour. Ihori et al. also noted a possible circular or feedback effect, in that frequent prosocial behaviour was also linked to a greater exposure to prosocial video game scenes.

This bidirectional relationship between exposure to elements of prosociality and prosocial behaviour was also found in a longitudinal study of 780 Japanese fifth graders (Gentile et el. 2009). Similar to Ihori et al.'s (2007) study, Gentile et al. assessed children's gaming habits and prosocial behaviours twice, with a 3- to 4-month time span between assessments. In both the initial and follow-up surveys, participants rated the frequency with which they had played video games with prosocial scenes during the past month. Prosocial scenes were defined as either “scenes is which characters help troubled persons” or “scenes where friendships or affections between parents and children are shown” (p. 757). Prosocial behaviour was measured as a function of how often the child had done each of four helpful behaviors in the past month. A structural equation modeling analysis revealed a significant, causal path from

the amount of prosocial game play at Time 1 to the amount of prosocial behaviour at Time 2 (3- to 4- months later). Similar to Ihori et al.'s conclusion, Gentile et al. noted that, "[t]his pattern of findings supports an upward spiral of prosocial gaming and behaviour (p. 757).

THE THREE CS: CONTENT, CONTEXT AND COOPERATION

The studies we have reviewed thus far have, for the most part, compared the effects of playing a prosocial video game to playing either a neutral or violent video game. However, prosocial effects are not limited to playing prosocial games.

Interestingly, prosocial effects have also been found in response to playing relaxing or neutral games. For example, Whitaker and Bushman (2012) examined the prosocial effects of playing either a neutral (*Super Mario Galaxy*, *Wii Sports Resort*), relaxing (*Endless Ocean*, the fishing mini-game in *The Legend of Zelda*, *Twilight Princess*), or violent video game (*Resident Evil 4*, *No More Heroes*). Undergraduates (N = 150) were randomly assigned to play one of the games for 20 minutes. Subsequent to playing the game, participants were paired with a "partner" (in reality, participants were competing against the computer) in a competitive reaction time task. Prior to each trial, participants set the level of an aversive noise that their "partner" would hear if their partner lost the trial; additionally, participants chose an amount of money that their partner would be awarded if their partner won the trial. Both prosocial behavior (i.e., the amount of monetary reward) and a reduction of aggressive behavior (i.e., the level of punishing noise) were measured. Compared to participants who played either the violent or neutral video game, participants who played the relaxing video game exhibited the least amount of aggressive behaviour. Moreover, participants who played either the neutral or relaxing video game were more generous in their monetary rewards than were participants who played the violent video game.

Beneficial and prosocial effects after playing neutral video games were also reported by Sestir and Bartholow (2010). They compared the effects of playing neutral, visual puzzle video games (*Zuma*, *The Next Tetris*) to the effects of playing violent video games (*Quake 3*, *Unreal Tournament*) and a no-game control condition (N = 111). Not only did participants who had played 30 minutes of a neutral video game generate significantly more prosocial responses on both a word-completion and a story-completion task than did participants who had played the violent video game, but the neutral-game participants also generated significantly more prosocial responses on both tasks than did participants in the no-game control condition. Furthermore, neutral-game participants displayed fewer aggressive responses on the story-completion task than did either the no-game participants or the violent-game participants. This was an important and novel finding, demonstrating that playing neutral video games, with no overtly prosocial content, can suppress aggressive cognitions in addition to elevating access to prosocial cognitions.

"Content is King" (meaning that content dictates effects) has been an adage long applied to the effects of playing video games. This has led to strongly held beliefs that violent video games will, assuredly, yield violent effects. If one extends this line of thinking, then neutral games will, assuredly, yield neutral effects. However, the results of both Whitaker and Bushman's (2012) and Sestir and Bartholow's (2010) studies indicate that content does not necessarily preordain outcome. The effects of playing any particular video game cannot

conclusively be determined based solely on its content. Since prosocial effects can arise from playing neutral video games, the question to ask then, is, "can prosocial effects arise from playing violent video games?".

In 2013, Ryan Robitaille tweeted "if content is king, then CONTEXT IS GOD". Although we have (ironically) taken this quote out of the design and marketing context it was originally quipped in, with regard to possible beneficial effects of playing video games, the importance of context is just as instrumental as content. Depending on the social context within which the video game is played (e.g., solo, cooperative, or competitive), beneficial and prosocial effects have been found even in response to playing violent video games. Cooperative game play has emerged as a pivotal moderator of the effects of playing video games.

In addition to noting context outside of the video game, the context within the video game is also an important consideration when examining the effects on players subsequent to game play. As noted in our introduction, a clear demarcation cannot necessarily be made between what is a "violent" and what is a "prosocial" video game. Prosocial scenes and themes, wherein characters help troubled persons and/or cooperate to achieve a common good, occur within video games that may be categorized as "violent". Hence, for many games, a more appropriate question with regard to violent and prosocial content may be one of "degree" rather than of "kind" or "genre" of game. Researchers such as Ihori et al. (2007) and Gentile et al. (2009) have taken this into consideration. In their studies, which evidenced longitudinal prosocial effects of exposure to prosocial content in video games, results were based on the frequency of prosocial scenes within a variety of video games rather than on a predefined category of "prosocial" game.

Jerabeck and Ferguson (2013) also utilized this approach when they examined the influence of violent video game play on beneficial behaviour. Seeking to examine the "interaction of content and context, specifically violent (antisocial and prosocial) content played cooperatively as a team" (p. 2575), 100 undergraduates were randomly assigned to play one of three games either cooperatively (paired with another participant) or solo. Games used were a violent antisocial game (*Borderlands*), a violent prosocial game (*Lego Star Wars III*), and a nonviolent, puzzle game (*Portal II*). Participants played their respective games for 45 minutes, and then engaged in a task designed (unbeknown to the participants) to measure aggressive behaviour. Participants were asked to select one of four hot sauces for another participant to taste as a (purportedly) unrelated taste-test study. The hot sauces ranged in intensity from mild to extremely hot. This measure of aggression was designed to be particularly salient, in that participants knew, and had met, who the recipient of their choice of hot sauce would be. Regardless of the video game played—nonviolent, violent social, or violent antisocial—participants who had engaged in cooperative video game play displayed significantly lower aggressive behaviour (as measured by intensity of hot sauce chosen for another to consume) than did participants who had engaged in solo video game play.

In addition to the beneficial effect of lowering subsequent aggressive behaviour, cooperative game play of violent video games has been demonstrated to increase subsequent prosocial behaviour. For example, in two experiments (Greitemeyer, Traut-Mattausch, & Osswald, 2012) German undergraduates (*N*s = 32, 70), who had played a violent video game (*Far Cry* in Study 1, *FlatOut* in Study 2*)* cooperatively, displayed more prosocial behaviour than did those who had played the same violent video game solo. Prosocial behaviour was measured the same way in both studies—via a give-some social dilemma. After playing the

video game (either cooperatively or alone), participants were informed that they were being paired with another participant from the same study who had been playing the video game in the room next door. Each participant was given four chips that were redeemable for 1 Euro each if the participant themselves redeemed the chip, but were worth 2 Euros if their partner redeemed the chips. Thus, if both participants chose to leave all of their chips for their partner to redeem, each participant was able earn 8 Euros. Across both studies, participants who had engaged in cooperative video game play displayed greater cooperate behaviour by choosing to leave more chips for their partner to redeem.

For Study 2, after engaging in the give-some social dilemma task, participants were asked to rate on a scale from 1 to 5, their feelings of cohesion—how much of a bond they felt with their "Euro chip" partner—and to what extent they trusted their "Euro chip" partner. Compared to participants who had played the violent video game alone, participants who had played the violent video game cooperatively reported significantly higher levels of cohesion and higher levels of trust with their subsequent give-some "Euro chip" partner. Mediational anlayses on these data examined possible pathways of causal effect leading to the positive relationship between cooperative play of violent video games and subsequent prosocial cooperative behaviour outside of the video game itself. Results identified a causal pathway whereby cooperative game play increased feelings of closeness and cohesion, which activated trust norms, which in turn evoked an increase in cooperative behaviour.

Cooperative game play is not just of concern to psychologists studying the effects of playing video games. The opportunity to engage in cooperative play is important to gamers themselves; in a survey conducted by El-Nasr et al. (2010), 77% of gamers indicated a preference for playing video games that had a cooperative player mode. The game industry is aware of this, and has begun to develop and produce top-rated video games—in particular violent video games—that can be played not only in competitive or single player modes, but also in cooperative player mode. Games such as *Resident Evil 5*, *Left4Dead*, *Halo*, and the immensely popular multi-player online game (MOG), *World of Warcraft* (El Nasr et al.; Ewoldsen et al., 2012, Lim & Lee, 2009) are just a few examples. Six patterns of cooperative design within video games have been discerned (Rocha, Mascarenhas, & Prada, 2008): 1) complementarity—players can assume different character roles to complement other players' activities; 2) synergies between abilities—a game character has the capacity to assist or alter another character's abilities; 3) abilities (such as healing) that cannot be used on one's own character, but can be used on others characters; 4) shared goals—patterns of play that require players to cooperate in order to advance the game; 5) special rules that further facilitate, encourage, and enhance cooperative play between gamers; and 6) synergies between goals—patterns within the game synchronize goals, thus, forcing players to cooperate. By employing these cooperative game play modes, researchers such as Ewoldsen et al. (2012) have replicated and extended Greitemeyer et al.'s (2012) results demonstrating that prosocial effects can result from playing violent video games cooperatively.

Ewoldsen et al. (2012) randomly assigned 119 experienced *Halo* (a violent video game) players to play the game for 15 minutes in one of three social contexts: direct competition (participants' task was to kill their opponent more times than they were killed), indirect competition (participants' task was to beat their opponent by advancing further through the game's single-player mode than their opponent did), or cooperative (participants' task was to advance as far as possible in the game by working together with their partner in the game's cooperative campaign mode which allows players to cooperatively battle computer-controlled

enemies). All participants were in separate rooms, competing or cooperating with the other players solely through the game.

After playing the violent video game, participants were again paired with their video game partner (or opponent, depending on the video game condition) to complete a series of 10 give-some social dilemma trials. In each trial, each participant was given four dimes which they could either keep or give to their partner. Dimes given away doubled in value for the partner. Decisions as to how many dimes to keep and how many dimes to give away were made privately, only being revealed after each player had made their decision for that round. This measure of prosocial behaviour assessed the extent to which each participant engaged in a tit-for-tat strategy, a pattern of behaviour indicative of future, long-term cooperative behaviour. Participants who had played the video game cooperatively used the tit-for-tat strategy significantly more than did participants who had played the video game in either of the competition modes. Moreover, participants who had played the violent video game cooperatively used the tit-for-tat strategy more than a fourth group of participants who had not played the violent video game until after they had engaged in the 10 trials of the give-some social dilemma task. Given the evidence supporting the effectiveness of the tit-for-tat strategy for maintaining long-term patterns of cooperative behaviour (Axelrod, 1984), we join Ewoldsen et al. in noting the importance of their study's results with regard to implications for prosocial effects of cooperative playing of video games of any genre. This is particularly so when coupled with findings such as Greitemeyer et al.'s (2012) which suggest that cooperative game play increases feelings of community and trust, character strengths which are highly valued and indicative of a wide variety of prosocial behaviours.

HARNESSING THE PROSOCIAL "SIDE EFFECTS" OF PLAYING VIDEO GAMES

As the research reviewed in this chapter highlights, the list of prosocial "side effects" of playing commercial video games designed for entertainment purposes is steadily growing. Innovative thinkers are now realizing the educational opportunities that are inherent within these "side effects". The "intersection of gaming and education" (American University, 2013) has opened the doors to a plethora of possibilities for using popular video games in unique ways within learning environments. Opening the doors even wider are games specifically designed to help players develop a variety of skills, such as those related to social interactions, leadership, and civic engagement, within diverse emotionally and cognitively engaging situations.

World of Warcraft and *Civilization*, two multi-player online games, illustrate the beneficial teaching experiences and opportunities that may be contained within massively popular video games (Bers 2010, Durga & Squire, 2009; Khoo, 2012). Khoo (2012) described the learning process that occurs within the virtual world of video games as an "inverted" process. She explained how in traditional models of learning, individuals learn *about* a topic or concept before embodying and putting these concepts into practice through interactions with others. Conversely, when individuals play video games, they "engage with the virtual worlds by learning 'to be', through immersion in their characters. This learning 'to be' is facilitated through the player's interactions with other players who make up the online

community in the game" (Khoo, p. 420). Players in games such as *World of Warcraft* participate in the game as members of online communities or "guilds"; although coming from all over the world, interaction occurs within the same virtual space. Ongoing online discussion forums within guilds are integral to the gaming experience.

To illustrate this inverted model of learning, Khoo (2012) provided several examples of how the moral dilemmas that players encounter within *World of Warcraft* facilitate the development of moral reasoning. For example, in one gaming scenario, plague-infested grain had been distributed to the villages in a particular kingdom. The Prince made the heartrending decision that all the villagers must be killed in order to stop the spread of the plague and save the rest of his kingdom. This story line spawned a great deal of earnest and thoughtful debate on this multifaceted moral dilemma. Merits of, and justifications for, the Prince's actions were pondered and debated. Questions were posed such as "was [the prince] acting on utilitarian principles (doing the greatest good for the greatest number)?".

Character and moral values are also developed during *World of Warcraft* play by more direct experiential means. Players can act as officers and managers of guilds. By assuming such civic roles, players gain experience in mediating and resolving disagreements and conflicts that occur within the game, and that extend outside of the game via the guild discussion forums (Khoo, 2012; Kahne, Middaugh, & Evans, 2009). Valuable lessons in leadership, governance, team building, organizational processes, social skills, and character education, can be learned via participation in the game and game-based discussions (Bers 2010; Khoo, 2012). For instance, choices must continuously be made within the game, and subsequently justified in public discussion forums, to serve one's self or others. One such choice involves using the game feature "Need, Greed, or Pass" with regard to loot acquired through raids. Khoo described how the guild expects more advanced players to "Pass" on loot so that other less experienced and less equipped players can gain by using the "Need" choice. Moreover, sanctions are swiftly instituted against players who use the game's "Greed" feature to make selfish choices.

Players recognize the ethical skills they acquire through their participation in online role-playing games. Empathy, an often difficult skill and character strength to teach and develop in children and adolescents, may be greatly enhanced by "playing a few alts". This game feature allows you to, as one *World of Warcraft* player noted, "see literally what wearing another's shoes provides" (Nordlinger, as cited in Poisso, 2010).

Another enormously popular multi-player online video game is *Civilization*. *Civilization,* an historical simulation game, is an exemplar of how popular video games can be adapted as teaching tools within an educational environment (Bers, 2010; Durga & Squire, 2009). Bers suggested that teachers can easily integrate *Civilization* into social studies, history, and government lessons. Doing so can not only help to deliver curriculum material in a lively and engaging manner, it can also help to contextualize the game's scenarios for players who may not always be aware of the actual sociocultural and political background scenarios which the game is based on. Furthermore, Bers noted that integration of *Civilization* into educational lessons can help students to understand the similarities and differences between video game virtual-world civic processes and real-life civic processes, thereby helping students to become more critical video game consumers while also carrying game lessons forward into real life.

Longitudinal research has demonstrated that, compared to conventional teaching methods, playing *Civilization* in a classroom context can help students learn historical names, places, and dates more effectively (Durga & Squire, 2009). Use of the game deeply enhanced

student engagement with history and social studies lessons—students used the game as a tool to explore alternate historical processes and events and to test hypotheses regarding current consequences of such alternate events. For example, Durga and Squire described how some students created simulated alternate contexts that examined possible conditions under which Native Americans could have held off European colonists. Advanced players developed models that allowed history to be viewed from different perspectives, and models that generated discussions on contemporary issues. Such detailed and integrative game modifications required the students to learn, understand, develop, and apply knowledge of historical and contemporary factors, such as rates of cultural expansion and how economic interdependence and organized religion have played (and continue to play) decisive roles in the conquests and colonization of civilizations. It is noteworthy that most of this knowledge was learned through active playing and strategizing within the game itself, and not via teacher lectures or instruction.

Video games are being developed which are designed specifically for teaching a diverse array of prosocial strengths, including empathy and civic engagement. For example, to facilitate empathy and social perspective taking, as part of the Social Aspects of Immersive Learning (SAIL) project, Harvard researchers developed a video game that enables players to take on different character's roles—in essence, a virtual experience of walking a mile in someone else's shoes (Schwartz, 2013). After experiencing situations as the opposing character in this game, participants were more likely to compromise in negotiations within the game compared to players who were provided with written information about the characters' perspectives.

Research teams from the Wisconsin Institute for Discovery (WID)'s Games+Learning+Society (GLS) group and the University of Wisconsin-Madison's Center for Investigating Healthy Minds are collaborating in the development of video games designed to teach middle-school-aged children prosocial behaviours such as recognizing others' emotions (an important aspect of empathy) (Spoon, 2013). In *Crystals of Kaydor*, players assume the role of a robot stranded on a foreign planet. To advance through the game, players must correctly gauge the emotions of the local aliens in a variety of social situations. Researchers will perform MRI scans on players of the video game before and after a two-week period of game play in order to gauge changes in brain circuits underlying social skills.

Recognizing emotions by identifying facial expressions is an important social skill that children with Autism Spectrum Disorders (ASD) experience great difficulty with. Researchers at the University of Victoria 's Brain and Cognition Lab and at the Yale Child Study Centre are developing video games designed to improve facial expression recognition abilities in children with ASD (University of Victoria, n.d.). Adults with Asperger's also struggle with these, and other, social skills. To address this, virtual world communities, such as *Brigadoon*, have been constructed where people with Asperger's can practice and enhance their social skills (Loftus, 2013). Brigadoon is a small, private community within the larger, public virtual world, *Second Life*. "Dooners", as inhabitants of Brigadoon have dubbed themselves, can interact with other Dooners at public meeting spaces and social events—situations that most would typically avoid in real-life. Being an active member of the Brigadoon community assists individuals with Asperger's in adapting to new situations and expanding beyond their natural preference for repetition and familiarity.

Video games that aim to increase prosocial civic engagement are helping extend social skills beyond individual social interactions. *Civic Seed*, for example, is being designed by

researchers at Tufts University (2014) to better prepare college students for community service. *Civic Seed* incorporates information and resources pertaining to local community service groups. Intertwined with the content are multi-level tasks that enable students to gain experience in activities such as organizational collaboration. Versions of *Civic Seed* that can be adapted for any institution and community are under development.

Advocacy groups have begun to identify and realize the potential that video games hold for galvanizing prosocial change. For example, human-rights activist and *New York Times* journalist, Nicholas Kristof, partnered with the non-profit group, Games for Change, mobile gaming company, Zynga, and numerous major corporations as funding investors, to create *Half the Sky Movement: The Game*, a Facebook video game spin-off from his book "Half the Sky: Turning Oppression into Opportunity for Women Worldwide", (Kristof & WuDunn, 2009). *Half the Sky Movement: The Game* familiarizes people with real-world issues and problems that women and girls face across the globe. This game raises not only awareness of social ills, it also raises funds to help combat these social injustices. One way that the game raises funds is through "hope bonds". Rather than waiting for their game "energy" to be restored after completing a quest, players can purchase an energy boost by buying hope bonds. The real-world money paid by players within this virtual-world is donated to a variety of charitable organizations working on behalf of women and children worldwide (Coffee, 2013; Half the Sky Movement: The Game, 2012).

Fundraising for real-world organizations has been taking place within virtual world video games for several years. For example, since 2005, the American Cancer Society (ACS) has held an annual Relay for Life fundraising event in *Second Life*, thereby tapping into the social consciences and goodwill of the more than15 million gamers who are citizens of the Second Life virtual world (American Cancer Society, 2014). The ACS has raised over $2 million in virtual-world fundraising events. Moreover, the ACS has established a virtual office in Second Life, where volunteers reach out to Second Life citizens who are seeking information and support in dealing with cancer in the real-world. Indeed, several large non-profit social change organizations (e.g., Amnesty International, Live and Learn in Kenya International), have established offices in Second Life from which they organize and host a variety of awareness and fund raising events that impact real-world citizens around the globe (Couturier, 2013).

MECHANISMS AND PATHWAYS: *HOW* VIDEO GAME PLAY CAUSES BEHAVIOURAL CHANGE

The research summarized in this chapter indicates that playing video games can promote prosocial behaviours. A variety of theoretical perspectives have been proposed to explain the mechanisms by which video games influence behaviour. In this section, we present a brief overview of the most commonly proposed mechanisms and pathways by which playing video games are thought to evoke prosocial behaviours and effects.

The General Learning Model predicts that video games can evoke both short-term and long-term effects via associations activated by a game's content (Buckley & Anderson, 2006). Prosocial games, wherein prosocial acts are necessary to advance in the video game, provide models of, and give directions for, prosocial behaviour. Immediate reinforcement for

prosocial behaviour is provided by the game, resulting in a feedback loop of continued prosocial behaviour. This loop primes, and rehearses, prosocial cognitive scripts that are subsequently activated outside of the video game. Gentile et al. (2009) suggested that the General Learning Model explains the increased prosocial behaviour demonstrated in studies where responses are tested shortly after playing prosocial video games (e.g., Greitemeyer & Osswald, 2010). The General Learning Model also predicts that repeatedly playing video games with prosocial content will result in long-term, prosocial changes to behaviour and traits through permanent changes in expectation schemata, attitudes, and affective traits such as empathy. Providing support for this prediction of the General Learning Model are longitudinal studies (e.g., Gentile et al., 2009; Ihori et al., 2007) that evidence a positive relationship between prosocial behaviours and/or traits and long-term play of video games with high prosocial content.

Drawing upon theories of moral education, character education, and care-ethics, Koo and Seider (2010) invoked the umbrella term "prosocial learning" to present pathways by which video games can evoke prosocial behaviours. Proponents of moral education adhere to a Kohlbergian (Kohlberg, 1981, 1984) perspective which claims that moral actions can be developed through reflecting on situations that invoke deep moral reasoning. Character education theorists advocate that morality and virtue are habits of behaviour formed through modeling, direct instruction, and practice (Ryan & Bohlin, 1999). An alternative to the character education approach is provided by care-ethics theorists (Noddings, 2002; Nussbaum, 2001) who position powerful literature as a primary means to foster virtue and empathy. Care-ethics proponents believe that immersing oneself in stories and novels enables individuals to identify with, and understand, others' viewpoints in a deeper, more impactful manner than instructional lectures or discussions can. Video games appear to facilitate prosocial behaviour by each of these pathways.

With regard to the moral education pathway, Koo and Seider (2010) provided examples of how educators can use scenes and rules from highly violent video games, such as *Grand Theft Auto IV*, as the basis of discussions centered on moral reflection and reasoning. Olson (2010) noted how, in focus group interviews of youths, a common theme emerged among the boys regarding the prosocial moral lessons that could be gleaned from the story lines of violent video games. The character education pathway of prosocial modeling, direct instruction, and practice is used when gamers play explicitly prosocial video games, such as *City Crisis* (e.g., Greitemeyer & Osswald, 2010), C*hibi Robo* (Saleem et al., 2012), or *Half the Sky Movement: The Game* (Coffee, 2013). Furthermore, video games that are not explicitly prosocial, yet still provide exposure to valuable character lessons and practice in exercising teambuilding and leadership skills, (e.g., *World of Warcraft*) also appear to take gamers along the character education pathway to prosocial behaviour (Barnett & Coulson, 2010; Bers, 2010; Khoo, 2012, Yee, 2006). Some video games appear to take gamers down all three pathways of prosocial learning. For example, in addition to utilizing the character education pathway of prosocial learning, *World of Warcraft* also appears to utilize the moral education and care-ethics pathways. We described earlier how Khoo (2012) used *World of Warcraft* to illustrate how the moral dilemmas which players may encounter in video games can help develop and facilitate moral reasoning (i.e., the moral education pathway). Poisso (2010) recounted how *World of Warcraft* players can gain empathy by immersing themselves in the intricate and ongoing sagas and chapters of the game's story by playing the game as more than one character (or alt), thus experiencing the story line from different viewpoints—

i.e., the care-ethics pathway (see also Jenkins, Clinton, Purushotma, Robison, & Weigel, 2006).

Theories of the adaptive and developmental functions of play may explain the positive, prosocial benefits of video game play (Granic et al., 2014), thus positioning gaming as part of normal, healthy development (Olson, 2010). Developmental psychologists such as Erikson (1977), Piaget (1962), and Gottman (1986) put forth that play, in general, and make-believe play in particular, allows individuals (especially children) to experiment with social roles and behaviours in a safe environment. Interactive play with others provides opportunities for individuals to explore emotional themes such as power, aggression, nurturance and growth. These experiences can help to resolve real-life emotional conflicts and foster emotional mastery. Video games provide unique opportunities to explore and experiment with a variety of behaviours, and to experience a variety of emotional and cognitive themes, within a multitude of diverse "safe" situations and spaces (Granic et al., 2014). Virtual communities, such as *Brigadoon* within *Second Life* (Loftus, 2013), and games such as *Civilization* (Bers, 2010), exemplify this mechanism by which video game play can help foster prosocial development and behaviour. Modifiable, multi-player games also provide safe "possibility spaces" (Squire 2008) within which boundaries can be tested and pushed, and acceptable limits of behaviour can be safely discovered (Barnett & Coulson, 2010; Chen, 2005; Kafai, Fields, & Giang, 2009; Küchlich, 2008; Searle & Kafai, 2009).

In addition to adaptive and developmental functions of play, Granic et al. (2014) proposed an additional mechanism by which playing video games may elicit beneficial effects. They suggested that because video games provide players with specific and immediate feedback regarding their actions and efforts within the game, video game play contributes to players acquiring an incremental theory of intelligence and skill. An incremental theory of intelligence postulates that intelligence (and skill) can be cultivated through effort and practice; this is in contrast to an entity theory of intelligence, which postulates that intelligence (and skill) is an innate, fixed trait with no room for improvement. Compared to individuals who hold an entity theory of intelligence, individuals who endorse an incremental theory of intelligence tend to persist longer in challenging activities (Dweck & Molden, 2005). Moreover, as Granic et al. noted, individuals who hold an incremental theory of intelligence view failure to achieve a desired goal as merely an indication that they need to bolster their efforts and remain engaged in the activity. Such a view of "failure" is predictive of better academic performance (Blackwell, Trzesniewski, & Dweck, 2007). To date, no studies have directly tested this theory regarding video game play leading to the acquisition of an incremental (versus entity) theory of intelligence and, in turn, increased persistence at real-world activities. However, Ventura, Shute, and Zhao (2013) demonstrated a positive relationship between extent of regular video game play and persistence at solving an anagram puzzle outside of the gaming context.

INTO THE FUTURE

A growing literature indicates that playing video games can provide a wide range of positive effects and benefits. As we have highlighted, playing prosocial video games can promote aspects of prosocial behaviours including increasing positive affect, generosity,

helping behaviour, and accessibility to prosocial thoughts, in addition to decreasing aggressive behaviour, hostile attribution bias, and accessibility to aggressive thoughts. Longitudinal research has supported these findings and demonstrated a positive feedback loop of prosocial gaming and prosocial behaviour. Even playing neutral video games can result in beneficial behaviours. Additionally, we have summarized research showing how valuable experience in moral reasoning, team building, and leadership can be gained from multi-player online games.

Researchers are just beginning to appreciate the intricacies of the relationship between video game playing and prosocial behaviour. Findings regarding the prosocial benefits of playing violent video games cooperatively, are one such example. Continued research examining moderators and mediators of the relationship between video game playing and prosocial behaviour is recommended. For example, given that personalizing one's avatar results in greater identification (Trepte & Reinecke, 2012) and can influence a player's behaviour even after the game is complete (Fischer, Kastenmuller, & Greitemeyer, 2010; Fox & Bailenson, 2009; Suh, Kim, & Suh, 2011), research examining avatar-gamer similarity as a moderator of the relationship between prosocial gaming and post-game prosocial behaviour is warranted. Research examining the physical characteristics of avatars has demonstrated a differential effect on the height and attractiveness of a gamer's avatar and subsequent off-line behaviour in a negotiation exercise (Yee, Bailenson, & Ducheneaut, 2009). Therefore, a viable extension of this research would be to examine the interaction between the physical characteristics of video game characters and avatars and the prosocial content of the video game on post-game prosocial behaviour.

Granic et al., (2014) suggested that future research should explore the relationship between playing video games and subsequent acquisition of theories of intelligence and increased persistence. They described several interesting future research directions including experimental studies to assess possible causal effects of video game play on the enhancement of real-world creativity, and studies which utilize temporally sensitive measures to strengthen our understanding of the relationships between positive mood and regular video game play. Other researchers have noted the need to further examine cross-cultural differences in the effects of playing prosocial video games (Koo & Seider, 2010). Additionally, although an astounding number and variety of video games are currently available, there are relatively few games based on positive constructs such as “hope”, “love”, “sacrifice”, “truth”, or “justice” (Rusch and Weise, 2008).

Both Johnson et al. (2013) and Ceranoglu (2010) have called for future research exploring the benefits of using video games within therapeutic settings. Although some research has already been conducted in this area (see Ceranoglu, 2010 for a review; Hull, 2009), more extensive research is required. We suggest that developing and testing video games based on validated positive psychology interventions would be fruitful. These interventions have been developed to enhance well-being (for reviews see Bolier et al., 2013; Magyar-Moe,2009; Sin & Lyubomirsky, 2009), and include include activities such as counting your blessings, practicing kindness, expressing gratitude, and engaging in philanthropic activities. Video games based on these well-being interventions would allow players to control avatars that can engage in behaviours empirically shown to increase positive well-being when performed by actual people. Post-game well-being could then be measured to assess the effectiveness of such video games. If successful, these games would have benefits beyond the immediate advantages to the individual player. Researchers could

use positive psychology-based video games as an efficient way to enhance well-being in order to study biological markers of happiness. Additionally, the games could be used in studies examining individual differences (e.g., in virtues, character strengths, self-concordant motivation) to determine whether naturally occurring behaviours, strengths, and inclinations mediate the efficacy of prosocial gaming. Furthermore, for clinicians and counselors, positive psychology-based video games, if effective, could provide a means to "jump start" positive affect to allow patients to have the motivation and sufficient positive affect to adhere to treatment plans.

The bulk of research studies examining the effects of video game play have, thus far, used console- or computer-based video games. Yet, 36% of gamers play video games on their smartphones, and 25% of gamers play video games on their wireless devices (ESA, 2013). These games include "app" type games, such as *Angry Birds* or *Bejeweled II*. Russoniello, O'Brien, and Parks (2009) demonstrated that playing app games that require a only short-term commitment, are highly accessible, and have minimal interfaces, can improve mood, promote relaxation, and help to buffer anxiety. The penetration of video games into our every day lives via these app-type games is almost certain to grow. Therefore, further research examining the prosocial effects of playing app-type video games on mobile devices is vital.

We hope that this chapter has led to a greater understanding of the numerous beneficial, prosocial effects that playing video games can have. We encourage collaboration between researchers and game developers to continue investigating this complex phenomenon, and to work together to explore the vast potential that gaming has for good.

APPENDIX

VIDEO GAME DESCRIPTIONS

Angry Birds: In the game, players use a slingshot to launch birds at pigs stationed on or within various structures, with the intent of destroying all the pigs on the playing field. This game is played on mobile devices such as smartphones. (Russoniello et al., 2009)

Bejeweled II: A puzzle game in which players swap gems to match jewels. This game is played on mobile devices such as smartphones. (Russoniello et al., 2009)

Borderlands: A first-person shooter game in which players play as bounty hunters killing on contract. (Jerabeck & Ferguson, 2013)

Chibi Robo: The goal of this game is to make your family happy by cleaning up, helping them out in their chores, and everyday tasks. As the player cleans up throughout the house, they earn Happy Points that improve their robot's ranking. (Gentile et al., 2009; Saleem et al., 2012)

City Crisis: The goal of the game is to save lives and to promote the security of a city. The player acts as a helicopter pilot who has to rescue citizens from burning houses, support the police by chasing burglars, and so on. (Greitemeyer & Osswald, 2010)

Civilization: This is a turn-based strategy game series in which you attempt to build an empire to stand the test of time. It begins in 4000 BC and the players attempt to expand and develop their empires through the ages from the ancient era until modern and near-future times. Players can become Ruler of the World by establishing and leading a civilization

from the dawn of man into the space age. Players can also wage war, conduct diplomacy, discover new technologies, go head-to-head with some of history's greatest leaders, and build the most powerful empire the world has ever known. (Bers 2010, Durga & Squire, 2009; Khoo, 2012)

Crash Twinsanity: In this game the player has to make his or her way across different stages, enemies, puzzles, and obstacles with Dr. Cortex (a computer-controlled character). The player has to fight and defeat numerous enemies along the way. (Gentile et al., 2009; Saleem et al., 2012)

Endless Ocean, the fishing mini-game in *The Legend of Zelda*: This game places the player in the role of a scuba diver exploring the Manaurai sea in search of sea life and sunken treasure. (Whitaker & Bushman, 2012)

Far Cry: A first-person shooter game; the main character is a lone mercenary. (Greitemeyer et al., 2012)

Firefighters: Saving Lives: The player saves game characters from burning houses. (Greitemeyer et al., 2012)

FlatOut: The player performs demolition derby deathmatches. (Greitemeyer et al., 2012)

Grand Theft Auto IV: This game is played from a third-person perspective in an open world environment, allowing the player to interact with the game world at their leisure. The single-player story follows Niko Bellic, a veteran of an unnamed war in Eastern Europe who comes to the United States in search of someone important, but quickly becomes entangled in a world of gangs, crime, and corruption. An online multiplayer mode is included with the game, allowing up to 16 players to engage in both co-operative and competitive gameplay in a recreation of the single-player setting. (Koo & Seider, 2010)

Halo: A first-person shooter game set in a military science fiction scenario. (Ewoldsen et al., 2012)

Lamers: This game is the aggressive version of Lemmings: All beings must be killed, and the goal is that no one reaches the exit. (Greitemeyer & Osswald, 2010)

Lego Star Wars III: Players fight bad guys and save the universe. (Jerabeck & Ferguson, 2013)

Lemmings: Players must guide groups of small beings through different worlds. The goal is to take care of the beings and to save them by leading them to the exit. (Greitemeyer & Osswald, 2009, 2010; Greitemeyer et al., 2012)

Mortal Kombat: A third-person fighting game. (Greitemeyer et al., 2012; Greitemeyer & Osswald, 2011)

No More Heroes: An action video game in which gameplay is open-ended, with the condition that the player must kill the top ten assassins to make the storyline progress. (Whitaker & Bushman, 2012)

Pinball: The goal of the game is to keep the ball on the pinball table by using the right and left triggers. (Greitemeyer et al., 2012)

Portal II: An adventure game involving the solving of puzzles without violent content. (Jerabeck & Ferguson, 2013)

Pure Pinball: The goal of the game is to keep the ball on the pinball table by using the right and left triggers. (Gentile et al., 2009; Saleem et al., 2012)

Quake 3: A first-person shooter game, where players run through a futuristic gladiator arena and must kill multiple enemies with a variety of weapons to succeed. (Sestir & Bartholow, 2010)

Resident Evil 4: A survival horror video game. The player controls the protagonist from a third-person, over-the-shoulder perspective in a mission to rescue the daughter of the President of the United States. The gameplay focuses on action and shootouts involving crowds of enemies in large open areas. (Whitaker & Bushman, 2012)

Super Mario Galaxy: To save Princess Peach and the universe, Mario travels from galaxy to galaxy in order to collect Power Stars, which are earned by completing levels in galaxies or defeating enemies. Each galaxy contains a number of planets and other space matter for the player to explore. (Whitaker & Bushman, 2012)

Super Mario Sunshine: Someone has polluted the island by leaving dirt marks and sludge everywhere. The player has to completely clean the area within each stage to progress to the next level. (Gentile et al., 2009)

S*uper Monkey Ball Deluxe*: The goal of the game is to advance through the different mazes by reaching the goal in the allocated time. The player must roll the monkey ball toward the goal, without allowing it to fall, within the time limit to advance to the next level. (Gentile et al., 2009; Saleem et al., 2012)

Tetris: A puzzle game where falling geometrical figures must be correctly positioned. (Greitemeyer & Osswald, 2009, 2010, 2011; Greitemeyer et al., 2012)

The Next Tetris: A visual puzzle game, in which players must manipulate objects into various patterns within a time limit in order to succeed (Sestir & Bartholow, 2010)

Twilight Princess: An action-adventure game focusing on combat, exploration, and item collection. (Whitaker & Bushman, 2012)

Ty2: The objective of this game is to get through the different stages and levels and get to Boss Cass before he takes over a country. On the way, the player (as Ty) has to fight with Cass's armed henchmen and robots that try to stop Ty. (Gentile et al., 2009; Saleem et al., 2012)

Unreal Tournament: A first-person shooter game, where players run through a futuristic gladiator arena and must kill multiple enemies with a variety of weapons to succeed. (Sestir & Bartholow, 2010)

Wii Sports Resort: A collection of sports games in which players mimic the actions performed in real life sports. (Whitaker & Bushman, 2012)

World of Warcraft: A massively multiplayer online role playing game. Players control a character avatar within a game world in third- or first-person view, exploring the landscape, fighting various monsters, completing quests, and interacting with non-player characters or other players. As characters become more developed, they gain various talents and skills, requiring the player to further define the abilities of that character. Characters can choose from a variety of professions. Much of the play involves completion of quests. While a character can be played on its own, players can group with others to tackle more challenging content. Most end-game challenges are designed in a way that they can only be overcome while in a group. In this way, character classes are used in specific roles within a group. (Bers 2010, Durga & Squire, 2009; Khoo, 2012)

Zuma: A visual puzzle game, in which players must manipulate objects into various patterns within a time limit in order to succeed (Sestir & Bartholow, 2010)

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